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program count_fermat_numbers
  implicit none
  integer :: n, max_n
  real(16) :: fermat_num
  logical :: is_prime
  Open(Unit=15, file='a1q5_in.txt')
  Open(Unit=17, file='a1q5_out.txt')

  Write(17,*) "Enter the maximum value of n:"
  read (15,*) max_n

  Write(17,*) "Fermat numbers are:"
  do n = 0, max_n

    fermat_num = 2.0**(2**n) + 1.0

    Write(17,*)"F(", n, ") = ", fermat_num

    is_prime=check_prime(fermat_num)
    if(is_prime)then
      Write(17,*)"F(", n, ") is a prime number"
    else
      Write(17,*)"F(", n, ") is not a prime number"
    end if
  end do

contains
  logical Function check_prime(num)
  Implicit none
  real(16)::num
  Intent(IN)::num
  real(16)::divisor
  check_prime=.True.

  if(num<=1)then
    check_prime=.false.
    return
  end if

  if(mod(num,2.0)==0 .AND. num /=2.0)then
    check_prime=.false.
    return
  end if

  do divisor=3,int(sqrt((num))),2
    if(Mod(num,divisor)==0)then
      check_prime=.false.
      exit
    end if
  end do

end function check_prime

end program count_fermat_numbers

```